

AIOps Assignment 1 - Q1

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Git Repository Link: <https://github.com/vrishabav/AIOps-A1>

Question 1

1.1

(a) Entanglement (Changing Anything Changes Everything) - the rounding logic on "estimated delivery time" was not supposed to be related to "favourite restaurants". However, the fact that the trained parameters of the recommendation model were built on differently rounded delivery times meant that they affected the favourite restaurants recommended. A seemingly external or superficial change changes the decision parameters of the model, and an "isolated" change causes a non-isolated effect.

(b) Undeclared Consumers - like explained in the slides, the marketing dashboard team has silently consumed a model's output via a shared table, creating invisible dependencies that block safe changes. This means that any change the ML team makes will not be communicated to or understood by the Marketing team. If one day the ML team decides to change something about their model - a parameter or a unit or an input column, the Marketing Team will not be aware of this until it is too late.

(c) Configuration and Glue-Code Debt - The 14 shell scripts are individual and importantly, undocumented. This means that for a person to effectively understand or make changes to the training pipeline, they need to understand everything about each of the 14 files, including which script produced what artefact in what format etc.

1.2

Mitigating the Configuration and Glue-Code Debt in (c): Firstly, the team should implement a systematic format for experiment tracking (eg: using MLFlow) along with a strict protocol to ensure reproducibility. Experiment tracking directly tackles Configuration debt, and allows there to be a "single source of truth" instead of several undocumented pipelines and manual steps being made. The training code should be instrumented with MLFlow so that all parameters, metrics and output artifacts are easily logged for each run. And to eliminate the reliance on scripts that aren't properly documented for setting up, an environment.yml should be set up to pin the execution environment which is reproducible using Conda/Mamba. Every experiment should also be tied to a specific Git commit hash. This ensures that any run's environment can be reproduced by any engineer or anyone working on the project, and there is no need to reverse-engineer undocumented commands or dependency issues.